Understanding the Impact of Screen Time on Sleep Quality

Project purpose

This study analyzes the hypothesis that "My quality of sleep decreases when my screen time increases'. The visualisation will aim to investigate the correlation between what I do with my screen before going to sleep and how I slept that day.

My primary research question in this project will be:

Does sleep quality decrease on nights when screen time before bed is higher than 60 minutes?

The question was based on my personal daily routine and my curiosity about whether or not I am sleeping better because of using my phone before going to sleep.

How the data was collected

In order to research this question, I gathered personal sleep-related information throughout a timeframe. The variables were the following and recorded on a daily basis:

- screen time before bed in minutes
- bedtime
- total sleep in hours
- sleep quality on a scale from 1 to 5

A rating of sleep quality was self-reported:

- **1** means poor sleep
- **5** means good sleep

I also recorded total sleep duration as well so that I could add another dimension of data into the visualization. This assisted me to not only consider the quality of sleep but also whether the individual slept 7 or more hours or less than 7 hours.

I decided to gather this data since it is relevant to my daily life. As screen time before sleeping is a daily routine, I wanted to experiment to see whether a pattern could be observed in my personal life rather than come to the conclusion.

Why this final design was chosen

The final visualization is a **scatterplot**I have chosen this format since a scatterplot is among the most understandable methods of demonstrating the connection between two numeric variables.

In this case:

- the **horizontal axis** demonstrates screen time before bed in minutes
- The vertical axis demonstrates the quality of sleep on a scale of 1 to 5.

It is easy to observe that the points on the graph will move downwards as the screen time increases.

I also drew a red vertical line at **60 minutes**. This line serves as a delineation and one can distinguish between the nights, which have fewer screen time and nights with more screen time. I selected 60 minutes as it provides a definite point of comparison to my research question and allows me to divide the image into two meaningful parts.

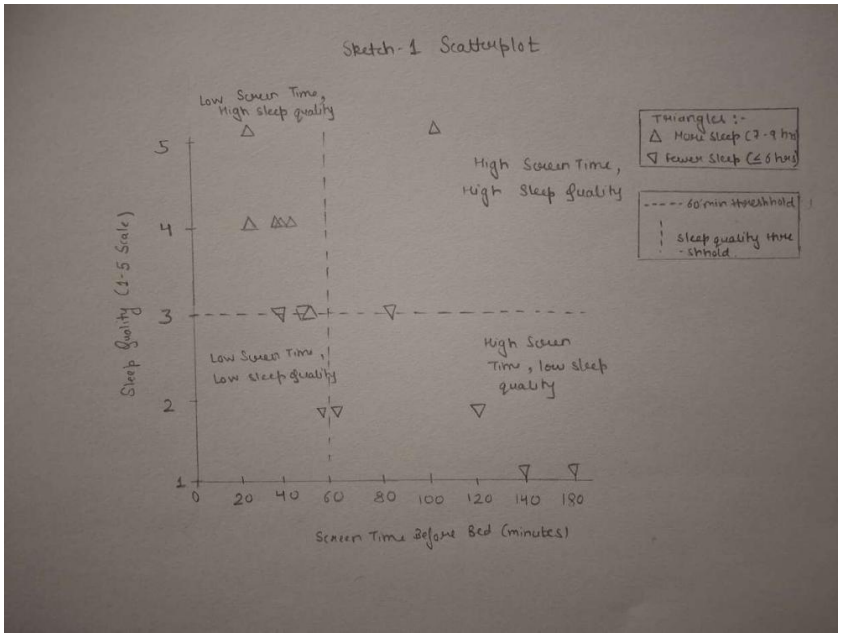
The reasons why this particular design was selected are that it is uncomplicated, easy to read, and directly addresses the central question of the project.

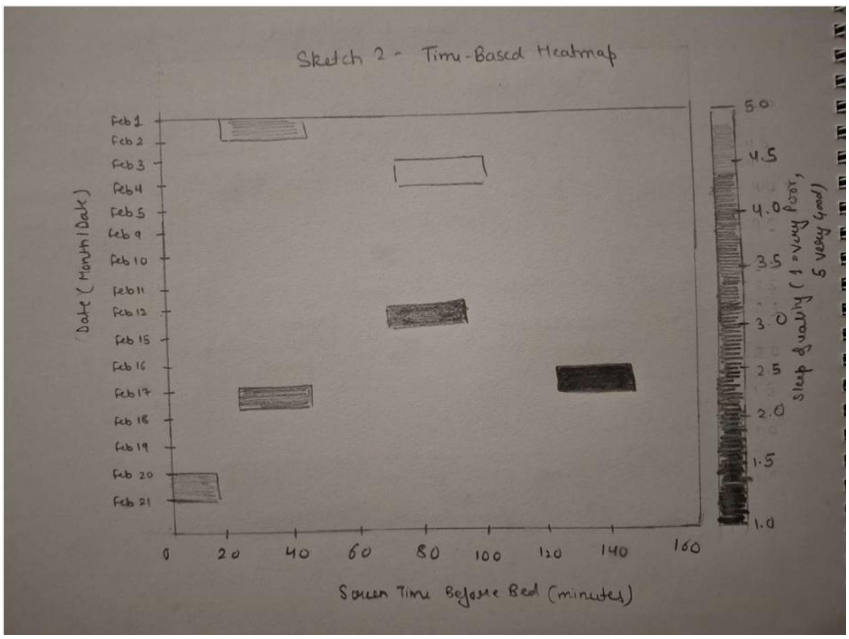
<u>Alternative Ideas and Why I Did Not Use Them</u>

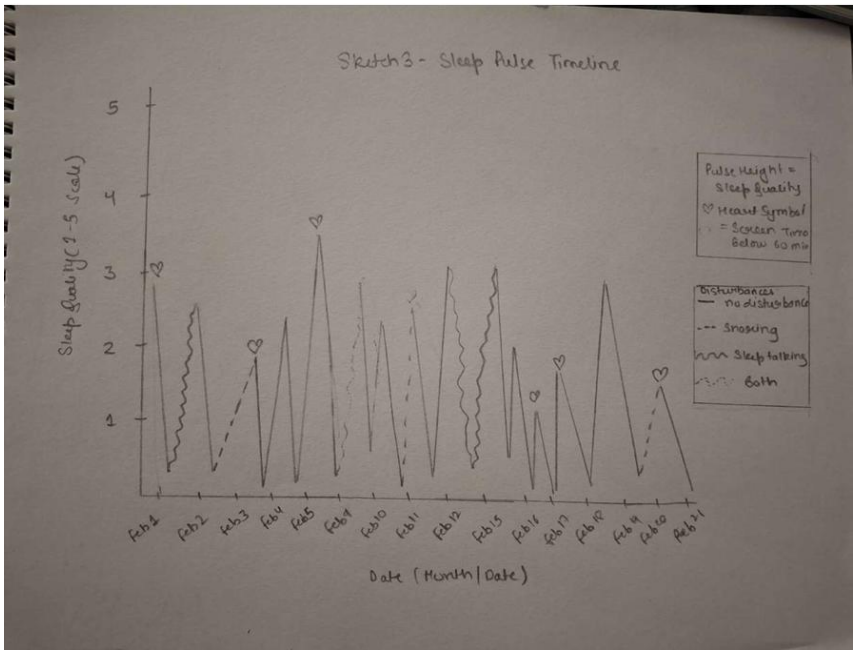
I experimented with other possible sketch ideas before deciding on this design.

One was to create a more elaborate multi-variable chart with more encodings. But then the design was complicated to read quickly.

A simplicity graph, which did not include threshold lines and the direction of the triangle, was another possible way, but it was not considered as informative in the visualization. It demonstrated the simplest association, although it was not that useful to understand how the duration of sleep relates to the same data.







How extra information is shown

Another visual variable that I added to the chart to make it more informative was triangle direction.

- Orange triangles up symbolize nights where there are 7 hours or more of sleep.
- Nights having less than 7 hours of sleep are indicated as downward blue triangles.

It does not present the correlation between the screen time and the sleep quality only, but also provides some idea as to whether longer sleep duration is more common with higher sleep quality.

To make the viewer get the picture within a short time, I added a legend on the right side. The threshold line of 60 minutes has been described in another legend box.

This was significant since the graph has more than just a single visual cue, and the legends simplify the process of interpretation of the visualization.

What the visualization shows

The scatterplot indicates that there is an overall trend wherein the quality of sleep would be better the lower the screen time before sleep.

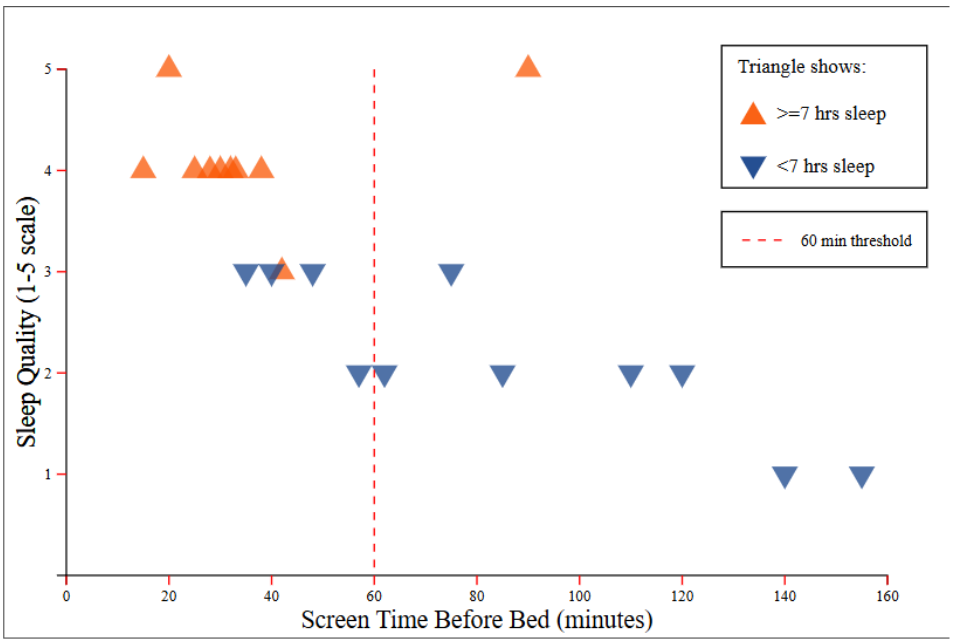
There are a lot of points on the left of the 60-minute threshold with sleep quality 3, 4, and 5. This implies that an average to good sleep quality can be attributed to low screen time.

There are more points on the right of the threshold with lower levels of sleep quality, including 1, 2, and 3. This implies that the more time children spend in front of the screen before going to bed, the more likely they are to have poor sleep habits.

The triangle directions as well indicate that most of the poor-quality sleep nights are related to less than 7 hours of sleep, which are depicted in downward blue triangles. At the same time, the fewer nights of higher quality are depicted by upward orange triangles, which implies 7 or more hours of sleep.

Simultaneously, the data is not consistently good. Some exceptions exist in which increased screen time still results in a higher quality of sleep. Due to this reason, the relationship is not absolute.

Nevertheless, the general trend of the graph proves the hypothesis that the increase in screen time prior to bed is largely linked with poorer sleep quality.



What I learned through my data

This project taught me that, based on my personal data, there is an apparent relationship between the screen time before sleep and sleep quality.

The bottom line is that lower screen time during evenings is more commonly associated with higher quality of sleep whereas higher screen time is more commonly associated with diminished sleep quality.

The other thing that I learned is that, there is more than one factor that causes sleep. Although there is a trend in the graph, the exceptions make me aware that the quality of sleep can also be related to such factors as stress, sleepiness, habit, or any other aspects of everyday life.

Therefore, the project does not demonstrate the only cause screen time is, but it demonstrates it is probably a significant factor.

My dataset:

DATE	SCREEN TIME (min)	BEDTIME	TOTAL SLEEP (hours)	SLEEP QUALITY (Very poor -> Poor; 1->5)
Feb 1	42	12:45 AM	7	3
Feb 2	15	11:50 PM	7.5	4
Feb 3	57	1:20 AM	6	2
Feb 4	35	12:30 AM	6	3
Feb 5	40	4:00 AM	3	3
Feb 6	Weekend (Friday night)			
Feb 7	Weekend (Saturday Night)			
Feb 8	Missed			
Feb 9	155	7:00 AM	4	1
Feb 10	48	12:40 AM	6.5	3
Feb 11	62	1:10 AM	6	2
Feb 12	28	12:05 AM	7	4
Feb 13	Weekend (Friday Night)			
Feb 14	Weekend (Saturday Night)			

Feb 15	33	12:20 AM	7	4
Feb 16	90	12:10 AM	9	5
Feb 17	120	4:00 AM	5	2
Feb 18	20	1:00 AM	8	5
Feb 19	75	3:00 AM	6	3
Feb 20	140	2:30 AM	4	1
Feb 21	30	1:15 AM	7	4
Feb 22	25	12:10 AM	7,5	4
Feb 23	85	2:10 AM	5.5	2
Feb 24	38	12:25 AM	7	4
Feb 25	110	3:20 AM	4.5	2
Feb 26	18	11:45 PM	8	5
Feb 27	Weekend (Friday Night)			
Feb 28	Weekend (Saturday Night)			
March 1	32	12:15 AM	7	4